# Estimating how much people influence their friends when friendship measurements are unreliable

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## Humans constantly influence their social contacts




## ...but measuring someone's friends is challenging.

If I ask you who your friends are:

- answers vary depending on how exactly I ask you,
- you'll probably forget day-to-day social contacts, and
- you might think someone is your friend, but they might disagree.

## We use estimated likelihood of friendship instead

$$Y_i = \beta_0 + W_i\beta_w + X_i\beta_x + \beta_y \underbrace{\sum_{j \neq i} \frac{A_{ij}}{d_i} Y_j}_{\text{peer contagion}} + \varepsilon_i \qquad \text{requires all } A_{ij}$$ 😬

$$Y_i = \theta_0 + \underbrace{W_i\theta_w}_{\substack{\text{observed} \\ \text{covariate} \\ \text{effects}}} + \underbrace{X_i\theta_x}_{\substack{\text{latent} \\ \text{positional} \\ \text{effect}}} + \theta_y \underbrace{\sum_{i \neq j} \frac{X_i^\top X_j}{\mathbb{E}[d_i \mid X]} Y_j}_{\text{latent contagion}} + \varepsilon_i \qquad \text{requires } \mathbb{P}(A_{ij} = 1)$$

## ...based on a low-rank network model



**Intuition**: stochastic blockmodels

Block indicators $Z_i$
Popularity parameters $\gamma_i$
Mixing matrix $B \in [0,1]^{d \times d}$

$$\mathbb{P}(A_{ij} = 1 \mid Z, \gamma, B) = \underbrace{\gamma_i Z_i B Z_j^\top \gamma_j}_{\text{rank } d}$$

Only assume undirected $A \in \mathbb{R}^{n \times n}$ with low rank $P = \mathbb{E}(A \mid X) = XX^\top$ and sub-gamma (Bernoulli, Poisson, Gaussian, Gamma, ...) deviations around $A$.

## Estimating contagion with noisy data

1. Humans constantly **influence each other**. We want to **estimate how strong social influence** is.
2. However, **measuring friendships is challenging**.
3. To estimate social influence, we **need to know who is friends with whom**.
4. We propose **a method that works with noisy friendship measurements**.

## ...to estimate how much people influence each other

Let $A \approx \hat{U}\hat{S}\hat{U}^\top$ be the rank $d$ SVD of $A$. Put $\hat{X} = \hat{U}\hat{S}^{1/2}$ and $\hat{P} = \hat{X}\hat{X}^\top$ and $\hat{d}_i = \sum_j \hat{P}_{ij}$ and $\hat{D} = \text{diag}(\hat{d}_1, ..., \hat{d}_n)$ and $\hat{G} = \hat{D}^{-1}\hat{P} \in \mathbb{R}^{n \times n}$. Let $\hat{Z} = [W\ \hat{X}\ \hat{G}y]$ and $\hat{H} = [W\ \hat{X}\ \hat{G}W\ \hat{G}\hat{X}\ \hat{G}^2W\ \hat{G}^2\hat{X}]$. Let $\hat{M} = \hat{H}(\hat{H}^\top\hat{H})^{-1}\hat{H}^\top$. Then

$$\hat{\theta}_{\text{TSLS}} = (\hat{Z}^\top\hat{M}\hat{Z})^{-1}\hat{Z}^\top\hat{M}y$$

**Theorem:** $\hat{\theta}_{\text{TSLS}}$ is asymptotically normal under the latent contagion model

$$\sqrt{n}\left(\hat{\theta}_{\text{TSLS}} - \theta\right) \to \mathcal{N}(0, \Sigma)$$

## This can be as good as knowing exact friendship!

**Theorem:** $\hat{\theta}_{\text{TSLS}}$ is asymptotically normal under the peer contagion model

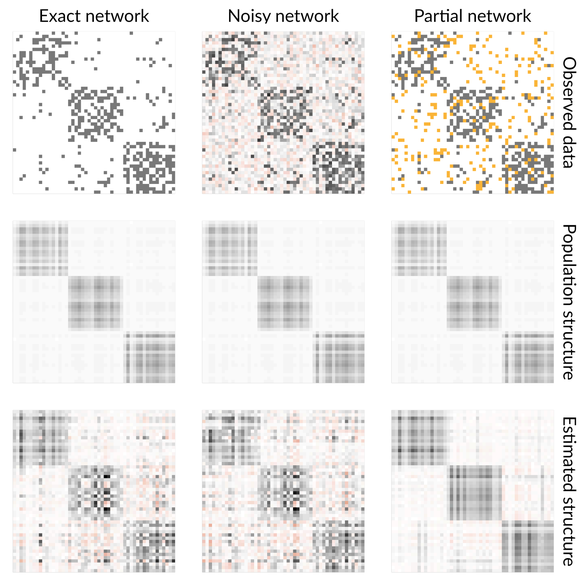
$$\sqrt{n}\left(\hat{\theta}_{\text{TSLS}} - \beta\right) \to \mathcal{N}(0, \Sigma)$$

How generally can we apply the strategy of replacing $A$ with $\mathbb{P}(A)$?

## More about Alex Hayes

- I'm an Assistant Professor in Statistics & Data Science
- I develop statistical methods for social networks and causal inference
- I write lots of `R` code and worked on some of the `tidyverse` packages
- I'm actively looking for collaborators!

## Our estimators can handle noisy measurements



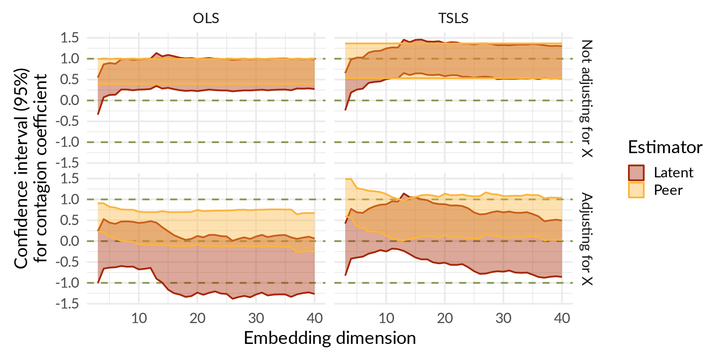

...because latent structure is estimable at fast rates even in noisy networks

## Anxiety before final exams amongst MBA students

Data on 241 MBA students from "Do Your Friends Stress You out? A Field Study of the Spread of Stress..." by Shihan et al (2023)

```
anxiety ~ personality + demographics + X + latent peer anxiety
anxiety ~ personality + demographics + X + peer anxiety
```

$X$ accounts for localized/group effects. Dimension of $X$ is unknown!




Model specification dramatically impacts inference